

Prathmesh Jain

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PROFESSIONAL SUMMARY

Full Stack AI Engineer with 1+ year of experience building GenAI-powered automation systems for BFSI workflows, specializing in agent orchestration, RAG pipelines, and deployable FastAPI + React applications.

EXPERIENCE

Tata Consultancy Services (TCS)

Oct 2024 - Present

Full Stack AI Engineer (Research & Innovation, CTO Team, BFSI Americas)

- **Agentic Compliance Screening System (KYC, AML, Adverse Media)** - Designed and implemented a GenAI-driven compliance screening system to automate KYC onboarding, sanctions checks, and adverse media analysis for BFSI workflows
- Built a stateful agentic workflow enabling iterative tool execution across Google search and Playwright-based web crawling exposed via **MCP**, sanctions datasets, and **RAG**-backed historical knowledge, reducing manual investigation time by 50%.
- Designed hybrid decision pipelines combining LLM reasoning with deterministic rules and structured outputs, reducing false positives during screening evaluations.
- Developed FastAPI-based services with retries, caching, fallbacks, and SSE streaming to support reliable execution of long-running AI workflows.
- **GenAI Payment Repair System** - Engineered a multi-stage AI pipeline to analyze and correct incomplete or inconsistent payment instructions derived from unstructured error descriptions
- Designed a multi-stage repair pipeline combining LLM-based semantic extraction, schema validation, and deterministic rule engines to automate resolution of common payment exception patterns without operator intervention.
- Built an agentic **CrewAI orchestration layer** with recursive repair and validation agents to resolve cascading field-level failures, thereby reducing rework from partial or unsafe repairs.
- Implemented FastAPI workflows with validation checkpoints, retries, fallbacks, and cost-aware LLM usage, cutting manual exception handling by 40% while maintaining correction accuracy.

PERSONAL PROJECTS

Patronix – E-commerce Platform | *React.js, Node.js, Express.js, MongoDB*

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- Built and deployed a full-stack e-commerce platform with live hosting, secure authentication, product catalog management, cart, payment gateway integration and order tracking flows.
- Optimized application performance and global state handling using Redux Toolkit.
- Designed modular, maintainable RESTful APIs with token-based authentication and clean architecture.

ExpenseGauge – Expense Tracking App | *React Native (Expo), Node.js, MongoDB*

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- Designed and shipped a cross-platform, offline-first expense management application with real users and a downloadable Android apk.
- Implemented optimistic UI updates and background sync queues to ensure seamless usage during network outages.
- Implemented secure authentication using short-lived access tokens and rotating refresh tokens for seamless session continuity.
- Developed admin dashboards for user management, transaction audits, and balance adjustments.
- Built role-based backend services supporting expense analytics with monthly trends and category-wise visualizations.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java

Backend: Node.js, Express.js, FastAPI, Flask

Frontend: React.js, Next.js, React Native, Redux Toolkit, Zustand, TailwindCSS

AI / LLM: LangChain, CrewAI, MCP, Prompt Engineering, RAG, Embeddings, Tool Calling

Database: MongoDB, PostgreSQL

Tools: Docker, Git, Postman, Playwright

EDUCATION

Vidyalankar Institute of Technology

Bachelors in Computer Engineering

GPA: 9.2 / 10

Mumbai, IN

May 2024